

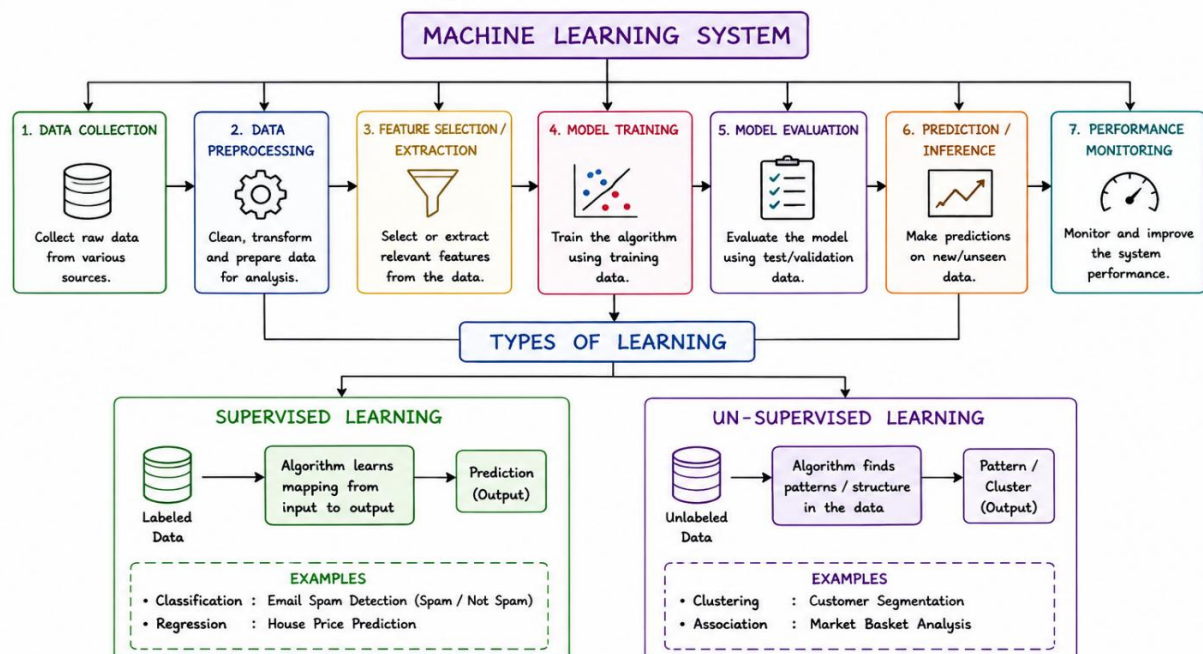
SIMATS ENGINEERING

ASSESSMENT TOOL 2

CONCEPT MAPPING

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

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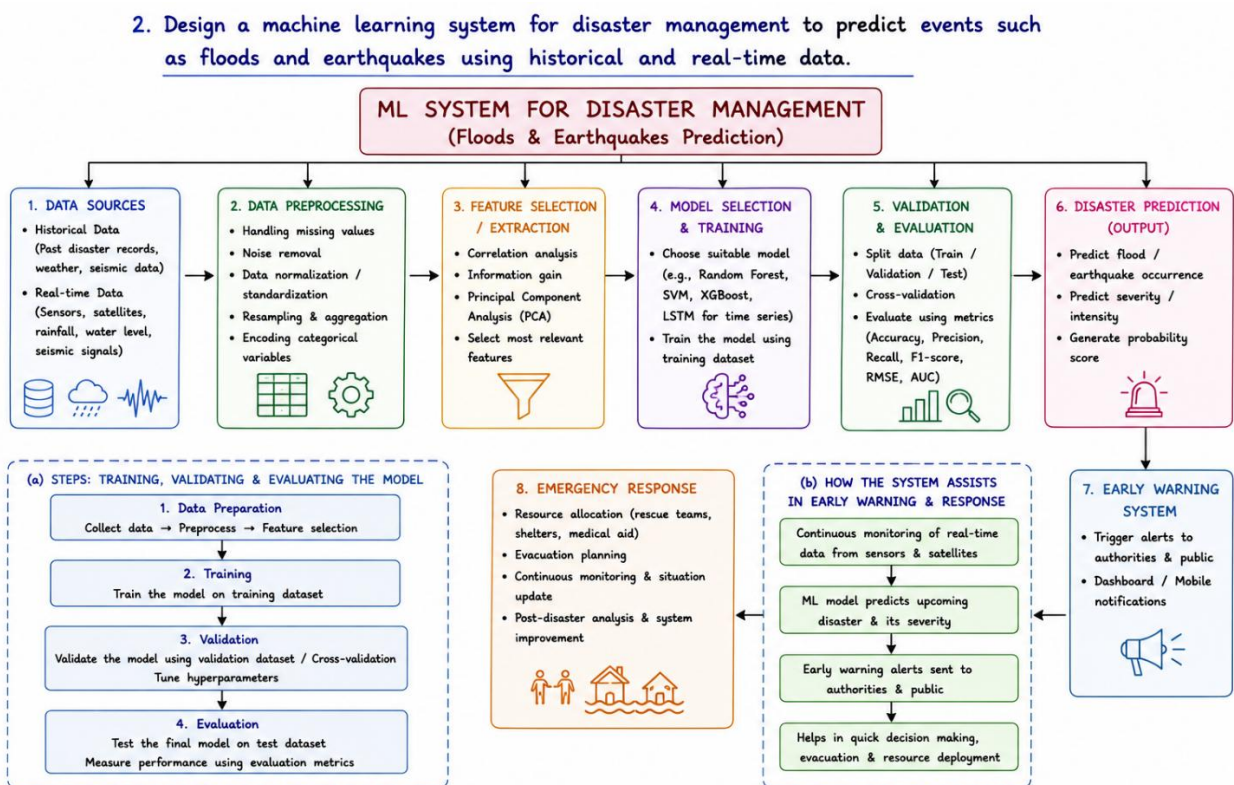
2. Design a machine learning system for disaster management to predict events such

as floods and earthquakes using historical and real-time data. Explain the choice of

model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response..



3. In the context of Find-S algorithm, consider the following dataset used to determine

whether a day is suitable for playing tennis based on attributes such as Outlook,

Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the

maximally specific hypothesis. Show the step-by-step updates of the hypothesis after

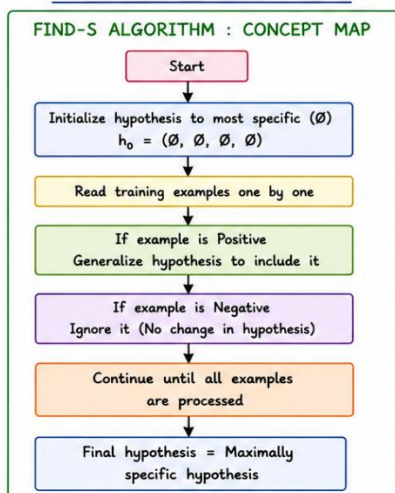
each positive training example and ignore all negative examples. Finally, provide the

final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

3. FIND-S ALGORITHM (Play Tennis)



Dataset :

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

STEP-BY-STEP FIND-S ALGORITHM (Consider Positive Examples Only)

Step	Training Example (Only Positive)	Example Details (Outlook, Temp, Humidity, Wind)	Hypothesis (h) After Update	Explanation
1	E ₃	(Overcast, Hot, High, Weak)	(Overcast, Hot, High, Weak)	Initialize to first positive example
2	E ₄	(Rainy, Mild, High, Weak)	(?, ?, High, Weak)	Generalize differing attributes
3	E ₅	(Rainy, Cool, Normal, Weak)	(?, ?, ?, Weak)	Generalize further
4	E ₇	(Overcast, Mild, Normal, Strong)	(?, ?, ?, ?)	Generalize all differing attributes

FINAL MAXIMALLY SPECIFIC HYPOTHESIS : $h = (?, ?, ?, ?)$

Note : All negative examples (E₁, E₂, E₆, E₈) are ignored.

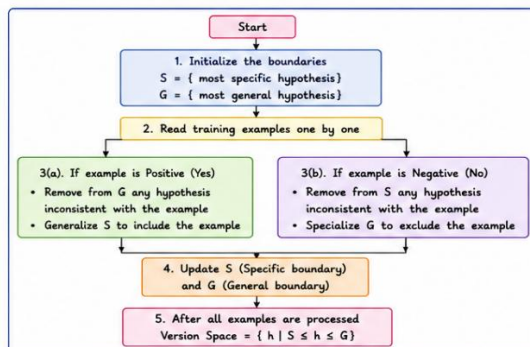
4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather,

Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

4. CANDIDATE ELIMINATION ALGORITHM : CONCEPT MAP



Dataset :

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Notation :

S → Specific boundary (set of most specific hypotheses)
G → General boundary (set of most general hypotheses)
? → Don't care (any value)

STEP-BY-STEP CANDIDATE ELIMINATION PROCESSING

Step	Training Example	Example Class	Update of S (Specific Boundary)				Update of G (General Boundary)				Explanation
			Weather	Temperature	Humidity	Wind	Weather	Temperature	Humidity	Wind	
0	-	-	∅	∅	∅	∅	?	?	?	?	Initial boundaries
1	(Sunny, Warm, Normal, Weak)	Yes	Sunny	Warm	Normal	Weak	?	?	?	?	First positive example Initialize S
2	(Sunny, Warm, High, Strong)	Yes	Sunny	Warm	?	?	?	?	?	?	Generalize S to cover this positive example
3	(Rainy, Cold, High, Weak)	No	Sunny	Warm	?	?	Sunny	?	?	?	Specialize G to exclude this negative example
4	(Sunny, Cold, Normal, Weak)	Yes	Sunny	?	?	Weak	Sunny	?	?	?	Generalize S to include this positive example
5	(Rainy, Warm, Normal, Strong)	No	Sunny	?	?	Weak	Sunny	?	Normal	?	Specialize G to exclude this negative example

FINAL VERSION SPACE (After all examples)

S (Specific Boundary) = { Sunny, ?, ?, Weak }
G (General Boundary) = { Sunny, ?, Normal, ? }

Version Space

$S \leq h \leq G$

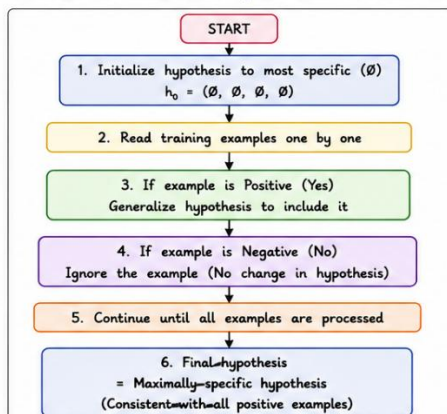
5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and

ignore negative examples. Finally, provide the final hypothesis.

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

5. FIND-S ALGORITHM (Customer Product Purchase) : CONCEPT MAP



DATASET :

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Attributes (in order) :

Age, Income, Student, Credit Rating

Note :

Consider only positive examples.
Ignore negative examples.

STEP-BY-STEP APPLICATION OF FIND-S ALGORITHM (Positive Examples Only)

Step	Training Example (Only Positive)	Example Details (Age, Income, Student, Credit Rating)	Current Hypothesis (h) After Update	Explanation
0	-	-	(∅, ∅, ∅, ∅)	Initialize to most specific hypothesis
1	E ₃	(Middle, High, No, Fair)	(Middle, High, No, Fair)	First positive example → hypothesis becomes this example
2	E ₄	(Senior, Medium, No, Fair)	(?, ?, No, Fair)	Generalize differing attributes (Age and Income differ)

How generalization works in Find-S :

- If attribute values are same → keep the same value
- If attribute values are different → replace with '?'
- We only generalize, never specialize.

FINAL MAXIMALLY SPECIFIC HYPOTHESIS

h = (?, ?, No, Fair)

Negative examples (E₁, E₂) are ignored in Find-S algorithm.

Interpretation :

The hypothesis says that a customer will buy the product if:

- Student = No
- Credit Rating = Fair
- Age and Income can be any value.

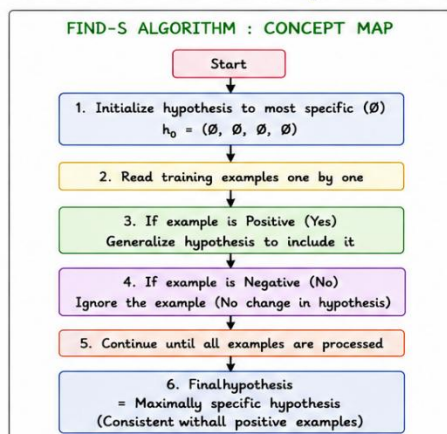
6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive

example only.

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

6. FIND-S ALGORITHM (Workout Prediction) : CONCEPT MAP



DATASET :

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

Attributes (in order) :

Weather, Temperature, Time, Mood

Note :

- Consider positive examples only.
- Ignore negative examples.
- Generalize only when attribute values differ.

STEP-BY-STEP APPLICATION OF FIND-S ALGORITHM (Positive Examples Only)

Step	Training Example (Only Positive)	Example Details (Weather, Temperature, Time, Mood)	Hypothesis (h) After Update	Explanation
0	-	-	(∅, ∅, ∅, ∅)	Initialize to most specific hypothesis
1	E ₂	(Rainy, Mild, Evening, Happy)	(Rainy, Mild, Evening, Happy)	First positive example → hypothesis becomes this example
2	E ₃	(Sunny, Mild, Evening, Happy)	(?, Mild, Evening, Happy)	Generalize differing attribute (Weather differs)
3	E ₄	(Cloudy, Cool, Morning, Happy)	(?, ?, ?, Happy)	Generalize differing attributes (Temperature and Time differ)

FINAL MAXIMALLY SPECIFIC HYPOTHESIS

h = (?, ?, ?, Happy)

Interpretation :

The hypothesis says a person will go for workout if Mood = Happy, regardless of Weather, Temperature and Time.

Summary :

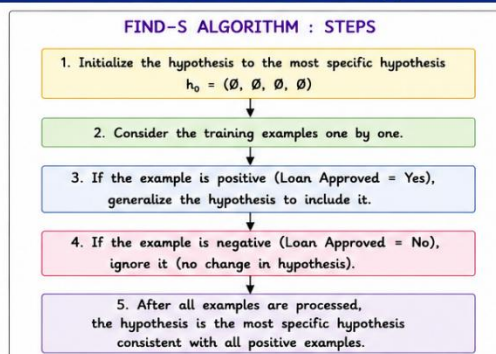
- Negative example (E₁) is ignored.
- Hypothesis is generalized only with positive examples.
- "?" means "any value" for that attribute.

7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

7. FIND-S ALGORITHM (Loan Approval)



DATASET

Example	Employment Status	Salary Level	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Attributes (in order):

Employment Status, Salary Level, Credit Score, Loan History
Here, $\emptyset$ represents the most specific value (no value).

Note:

- Consider only positive examples (Yes).
- Ignore negative examples (No).

STEP-BY-STEP APPLICATION OF FIND-S ALGORITHM

Step	Training Example (Positive Only)	Loan Approved	Hypothesis (h) After Update				Explanation
			Employment Status	Salary Level	Credit Score	Loan History	
0	-	-	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	Initialize with the most specific hypothesis.
1	Example 1	Yes	Employed	High	Good	Clean	First positive example. Generalize h_0 to include Example 1.
2	Example 2	Yes	Employed	?	Good	Clean	Salary differs (High vs Medium), so generalize that attribute to '?'. No change in hypothesis.
3	Example 3	No	Employed	?	Good	Clean	Negative example → Ignore. No change in hypothesis.
4	Example 4	Yes	Employed	?	?	Clean	Credit Score differs (Good vs Excellent), so generalize that attribute to '?'. No change in hypothesis.

FINAL HYPOTHESIS (Most Specific Hypothesis)

$h = (\text{Employed}, ?, ?, \text{Clean})$

This hypothesis is consistent with all positive examples.

INTERPRETATION

The hypothesis says: A loan will be approved if the applicant is Employed, has any Salary Level, any Credit Score, and has a Clean Loan History.

SUMMARY

- Positive examples: 1, 2, 4
- Negative example: 3 (ignored)
- Final hypothesis is the most specific hypothesis that covers all positive examples.

8. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example.

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

8. CANDIDATE ELIMINATION ALGORITHM (Customer Subscription)

PROBLEM

Given training examples, find the version space by computing the specific boundary (S) and general boundary (G) for classifying whether a customer will Subscribe (Yes/No).

ALGORITHM

- S : Set of maximally specific hypotheses (start with most specific hypothesis).
- G : Set of maximally general hypotheses (start with most general hypothesis).
- Update S and G for each training example.
- Version Space = { all hypotheses h such that $\exists s \in S$ and $g \in G$ where $s \leq h$ }

DATASET

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Attributes (in order):

Age Group, Income Level, Internet Usage, Device Type

Symbols:

∅ : Most specific value (null)
? : Most general value (any)

STEP-BY-STEP UPDATES OF S AND G

Step	Training Example (Subscribe)	Example Details (Age, Income, Internet Usage, Device)	Specific Boundary (S) After Update	General Boundary (G) After Update	Explanation
0 (Initial)	-	-	{ ∅, ∅, ∅, ∅ }	{ ?, ?, ?, ? }	Initialize S with most specific and G with most general.
1	E ₁ (Yes)	{ Young, High, Frequent, Mobile }	{ Young, High, Frequent, Mobile }	{ ?, ?, ?, ? }	Positive example: Make S more specific (to E ₁).
2	E ₂ (Yes)	{ Young, High, Occasional, Mobile }	{ Young, High, ?, Mobile }	{ ?, ?, ?, ? }	Positive example: Generalize differing attribute (Internet Usage).
3	E ₃ (No)	{ Middle, Low, Frequent, Desktop }	{ Young, High, ?, Mobile }	{ Young, ?, ?, ? }	Negative example: Remove hypotheses in G that cover E ₃ (G becomes more specific).
4	E ₄ (Yes)	{ Young, Medium, Frequent, Mobile }	{ Young, ?, ?, Mobile }	{ Young, ?, ?, ? }	Positive example: Generalize Income (High vs Medium) and Internet Usage.
5	E ₅ (No)	{ Senior, High, Occasional, Desktop }	{ Young, ?, ?, Mobile }	{ Young, ?, ?, ? }	Negative example: G already excludes E ₅ (different Age and Device), so no change.

FINAL VERSION SPACE

Specific Boundary (S) : { { Young, ?, ?, Mobile } }
General Boundary (G) : { { Young, ?, ?, ? } }

All hypotheses h such that { Young, ?, ?, Mobile } ≤_h ≤ { Young, ?, ?, ? } constitute the version space.

INTERPRETATION

A customer will Subscribe (Yes) if:

- Age Group = Young
- Device Type = Mobile
- Income Level and Internet Usage can be any.

SUMMARY

- After processing all training examples, S is the most specific hypothesis consistent with all positive examples.
- G is the most general set of hypotheses that exclude all negative examples while remaining general.
- The version space captures all valid hypotheses between S and G.

9. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the version space (S and G boundaries) and show the updates after each example.

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

9. CANDIDATE ELIMINATION ALGORITHM (Healthy Lifestyle)

PROBLEM

Given training examples, find the version space by computing the specific boundary (S) and general boundary (G) for classifying whether a patient follows a Healthy Lifestyle (Yes/No).

ALGORITHM (Candidate Elimination)

- S : Set of maximally specific hypotheses (start with most specific hypothesis).
 - G : Set of maximally general hypotheses (start with most general hypothesis).
 - Update S and G for each training example.
 - Version Space = $\{ h \mid \exists s \in S \text{ and } g \in G \text{ such that } s \leq h \leq g \}$
- Where:
 $s \leq h$ means s is more specific or equal to h
 $h \leq g$ means h is more general or equal to g

DATASET

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

Attributes (in order):

Diet, Exercise, Sleep, Stress Level

Symbols:

$\emptyset$: Most specific value (null)
 $?$: Most general value (any)

Note:

- Consider only positive examples (Yes)
- Negative examples help in generalizing G

STEP-BY-STEP UPDATES OF S AND G

Step	Training Example (Healthy Lifestyle)	Example Details (Diet, Exercise, Sleep, Stress)	Specific Boundary (S) After Update	General Boundary (G) After Update	Explanation
0 (Initial)	-	-	$(\emptyset, \emptyset, \emptyset, \emptyset)$	$(?, ?, ?, ?)$	Initialize S with most specific and G with most general.
1	E_1 (Yes)	$\langle \text{Balanced, Regular, Adequate, Low} \rangle$	$\langle \text{Balanced, Regular, Adequate, Low} \rangle$	$\langle ?, ?, ?, ? \rangle$	Positive example: Make S more specific (to E_1).
2	E_2 (Yes)	$\langle \text{Balanced, Regular, Poor, Medium} \rangle$	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\langle ?, ?, ?, ? \rangle$	Positive example: Generalize differing attributes (Sleep and Stress Level).
3	E_3 (No)	$\langle \text{Junk, None, Poor, High} \rangle$	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$	Negative example: Remove from G any hypothesis that covers E_3 . (Diet must be Balanced).
4	E_4 (Yes)	$\langle \text{Balanced, Moderate, Adequate, Low} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$	Positive example: Generalize Exercise (Regular vs Moderate), Sleep (? stays), Stress (? stays).
5	E_5 (No)	$\langle \text{Junk, Regular, Adequate, High} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$	Negative example: Already excluded since Diet = Junk ($\neq$ Balanced). No change in G .

FINAL VERSION SPACE

Specific Boundary (S) : $\{ \text{Balanced, ?, ?, ?} \}$
 General Boundary (G) : $\{ \text{Balanced, ?, ?, ?} \}$

All hypotheses h such that
 $\langle \text{Balanced, ?, ?, ?} \rangle \leq h \leq \langle \text{Balanced, ?, ?, ?} \rangle$
 constitute the version space.

INTERPRETATION

A patient is predicted to have a Healthy Lifestyle (Yes) if:

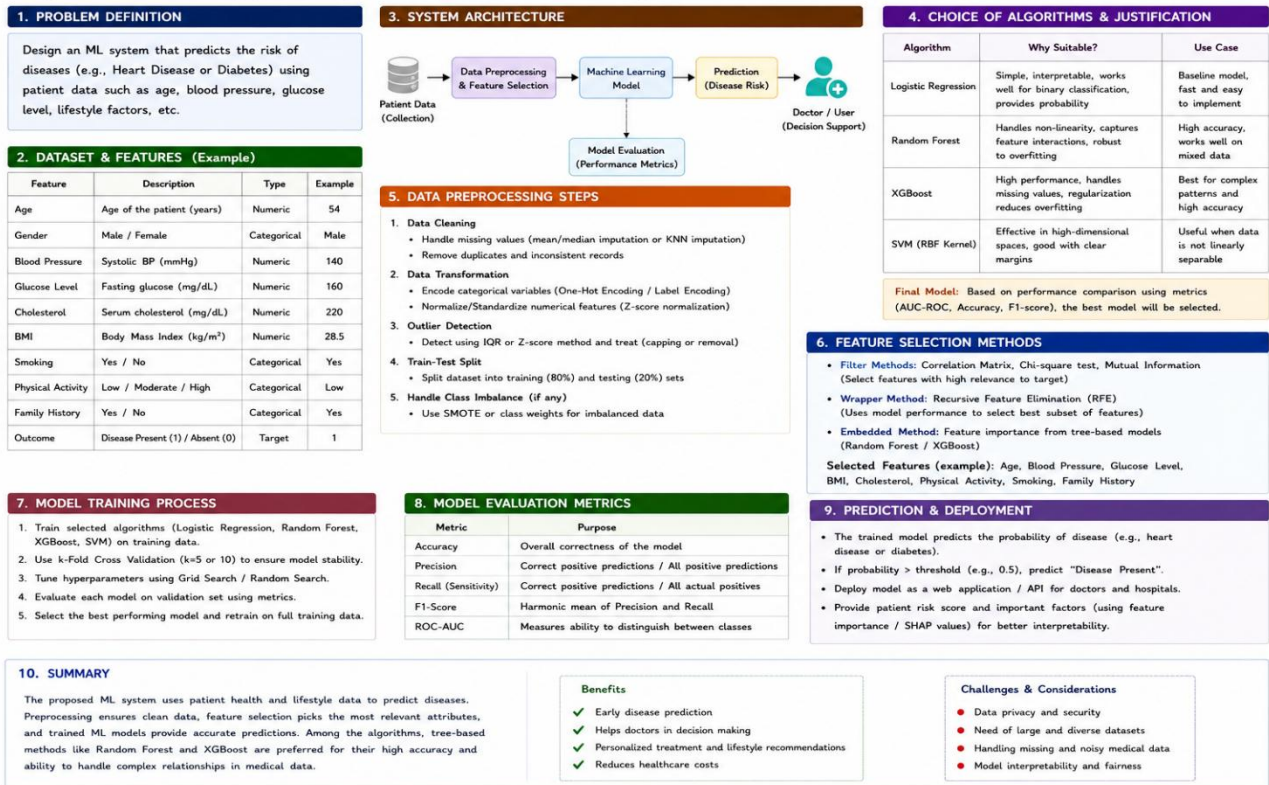
- Diet = Balanced
- Exercise can be any (Regular or Moderate or None)
- Sleep can be any (Adequate or Poor or None)
- Stress Level can be any (Low or Medium or High) (i.e., only Diet must be Balanced)

SUMMARY

- After processing all training examples,
 $S = \{ \text{Balanced, ?, ?, ?} \}$ (most specific)
 $G = \{ \text{Balanced, ?, ?, ?} \}$ (most general)
- The version space includes all hypotheses where Diet = Balanced and other attributes can take any value.
- This space captures all hypotheses consistent with the data.

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions

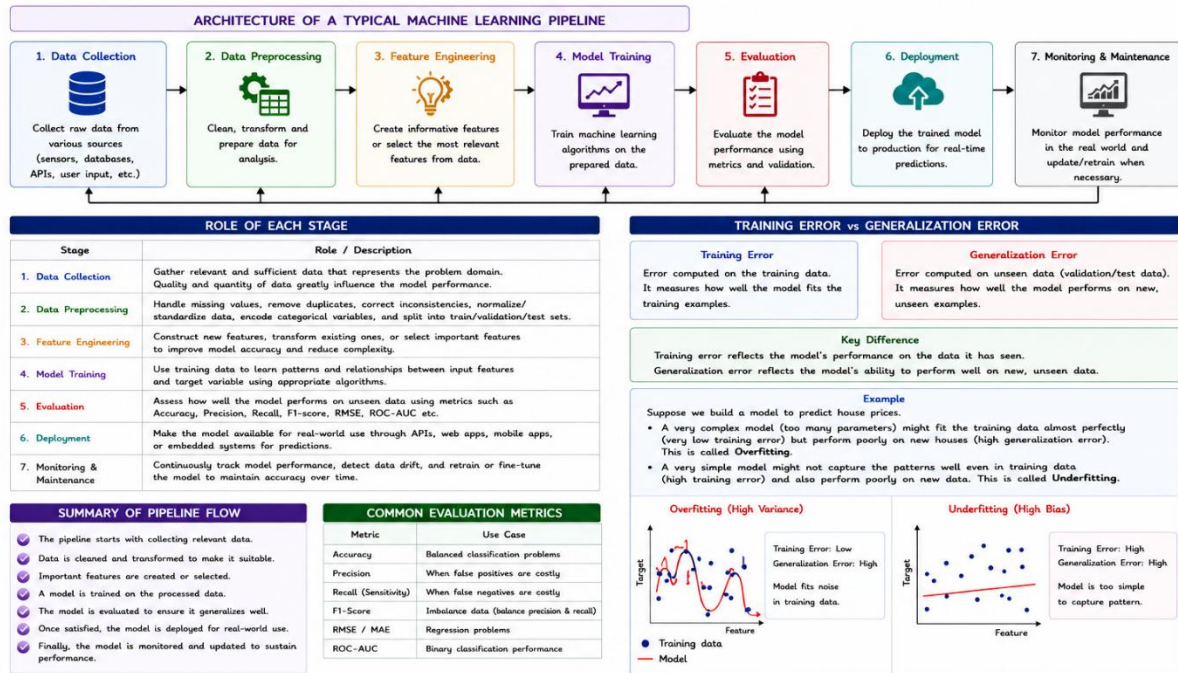
10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors.



11. Explain the architecture of a typical machine learning pipeline with a neat diagram.

Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.



12. Design a machine learning model for spam email classification using labeled data.

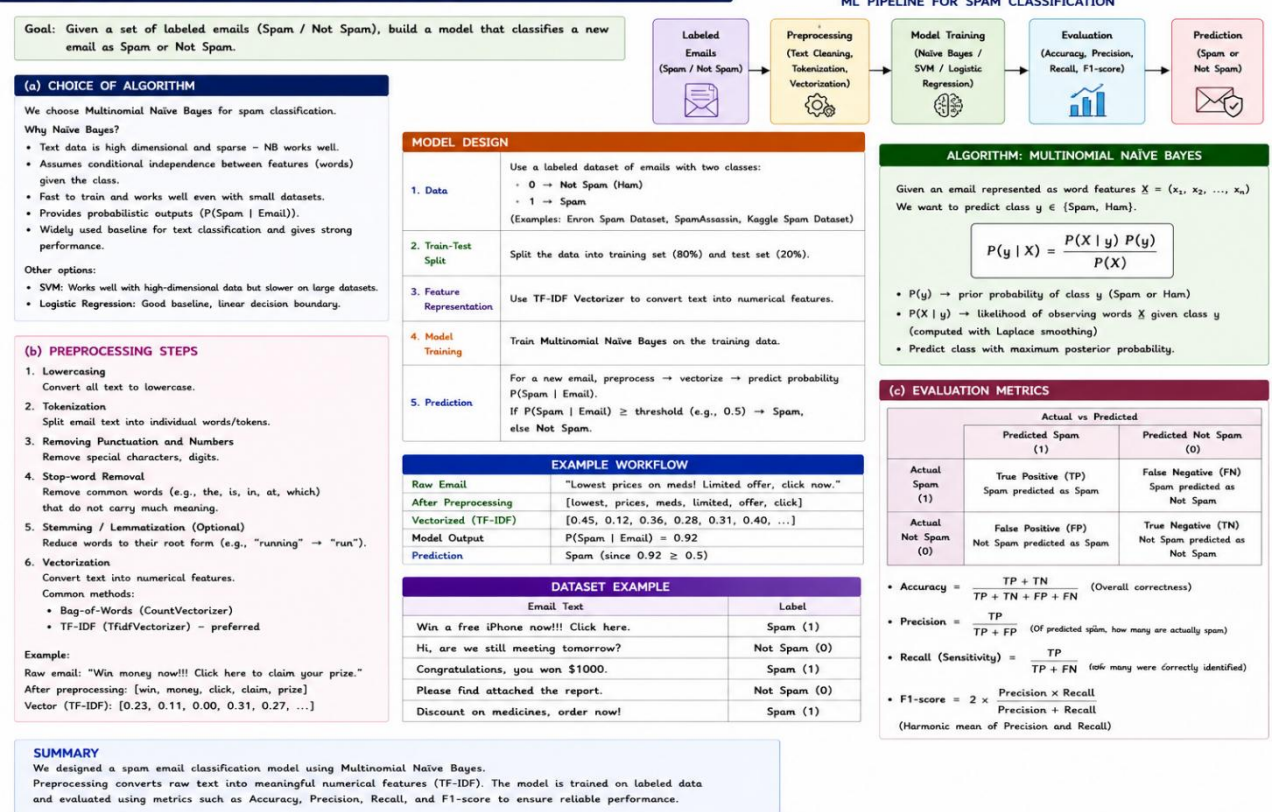
(5 Marks)

(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).

(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.

(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

12. Design a Machine Learning Model for Spam Email Classification



13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes:

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.

(a) Explain the initialization of S (specific boundary) and G (general boundary).

(b) Show how S and G change with each training example.

(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes:

Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.

Training Examples:

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
E1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
E2	Sunny	Warm	High	Strong	Warm	Same	Yes
E3	Rainy	Cold	High	Strong	Warm	Change	No
E4	Sunny	Warm	High	Strong	Cool	Change	Yes
E5	Sunny	Cold	Normal	Strong	Cool	Change	No
E6	Rainy	Warm	Normal	Strong	Warm	Same	Yes

(a) Initialization of S and G

- S (Specific Boundary): Most specific hypothesis. Initially it contains only the most specific value ($\emptyset$) for every attribute.

$$S_0 = \{ \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle \}$$

- G (General Boundary): Most general hypothesis. Initially it contains only the most general value (?) for every attribute.

$$G_0 = \{ \langle ?, ?, ?, ?, ?, ?, ? \rangle \}$$

Where,

$\emptyset$ = most specific value (matches only a single value)

? = most general value (matches any value)

(a) Explain the initialization of S (specific boundary) and G (general boundary).

(b) Step-by-step changes in S and G with each training example

Step	Training Example	EnjoySport	Action	S (Specific Boundary)	G (General Boundary)	Explanation
0 (Initial)	-	-	Initialization	$\langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$	$\langle ?, ?, ?, ?, ?, ?, ? \rangle$	Start with most specific S and most general G.
1	E1	Yes	Positive example: Make S more specific.	$\langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$	$\langle ?, ?, ?, ?, ?, ?, ? \rangle$	S becomes E1; G unchanged.
2	E2	Yes	Positive example: Generalize S where needed.	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\langle ?, ?, ?, ?, ?, ?, ? \rangle$	Humidity differs (Normal vs High) → replace with ?
3	E3	No	Negative example: Generalize G to exclude E3.	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?, ?} \rangle, \langle ?, ?, ?, ?, ?, \text{Same} \rangle \}$	All hypotheses in G that cover E3 are specialized minimally to exclude it.
4	E4	Yes	Positive example: Minimize G and generalize S if possible.	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?, ?} \rangle \}$	S generalized at Water (Warm → ?), Forecast (Same → ?). G reduced (third member removed since it doesn't cover any positive).
5	E5	No	Negative example: Generalize G to exclude E5.	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?, ?} \rangle \}$	First member ruled out (covers E5), so specialize it on AirTemp. Second member unchanged.
6	E6	Yes	Positive example: Minimize G and generalize S if possible.	$\langle ?, \text{Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?, ?} \rangle, \langle \text{Rainy, Warm, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?, ?} \rangle \}$	S generalized on Sky (Sunny & Rainy → ?). G updated & minimized to include Rainy and maintain coverage of E6.

(c) Comparison: Candidate-Elimination vs Find-S

Aspect	Candidate-Elimination Algorithm	Find-S Algorithm
Boundary Representation	Maintains two boundaries: S (most specific) and G (most general).	Maintains only S (most specific).
Use of Negative Examples	Uses both positive and negative examples to refine hypothesis space.	Uses only positive examples.
Hypothesis Space	Maintains a version space (all hypotheses between S and G).	Returns a single hypothesis (may be over-specific).
Generalization Capability	Higher generalization; can find all possible consistent hypotheses.	Lower generalization; may miss more general hypotheses.
Output	Set of all consistent hypotheses (bounded by S and G).	Single most specific hypothesis consistent with positive examples.
Handling Ambiguity	Better handles ambiguity and noise using both types of examples.	Sensitive to noise and cannot rule out overly specific hypotheses.
Computational Complexity	Higher (maintains sets and updates both S and G).	Lower (updates only S).

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.

(a) Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).

(b) Explain feature scaling and handling missing data.

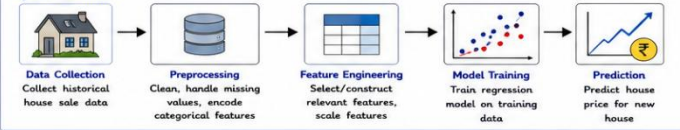
(c) Describe how overfitting can be detected and avoided.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

Problem Statement

Design an ML system to predict house prices (continuous target) using features: Location, Size (sq.ft), Number of Rooms, and Amenities.

System Overview



(a) Choice of Regression Model and Justification

Chosen Model: Random Forest Regressor

Justification:

- House price prediction is a complex non-linear problem with interactions between features.
- Random Forest is an ensemble of decision trees that can capture non-linear relationships without requiring feature scaling.
- It handles both numerical and categorical features well (after encoding).
- Robust to outliers and noise and has low variance compared to a single decision tree.
- Provides feature importance, which helps in understanding key drivers of price.
- Generally gives higher accuracy than simple Linear Regression and more stable than a single Decision Tree.

Model	Pros	Cons	When Suitable
Linear Regression	Simple, interpretable, fast	Assumes linearity, sensitive to outliers	When relationship is approximately linear
Decision Tree Regressor	Captures non-linear patterns, easy to interpret	High variance, prone to overfitting	Small datasets, when interpretability is critical
Random Forest Regressor	High accuracy, handles non-linearity, robust, feature importance	Less interpretable than single tree, more computation	Most real-world house price prediction problems

(b) Feature Scaling and Handling Missing Data

Feature Scaling

- Features like Size (in sq.ft) and Number of Rooms may have larger ranges than others.
- Scaling ensures all features contribute equally to the model and improves performance (important for Linear Regression and other distance-based models).
- Common Methods:
 - Standardization (Z-score scaling): $z = \frac{x - \mu}{\sigma}$
 - Min-Max Scaling: $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$
- Tree-based models (e.g., Random Forest) are less sensitive to scaling, but scaling is still good practice.

Handling Missing Data

- Missing values can degrade model performance.
- Strategies:
 - Numerical Features:
 - Impute with mean/median (simple and effective)
 - Or use KNN/Regression imputation
 - Categorical Features:
 - Impute with mode (most frequent)
 - If missingness is significant, add a missing indicator feature (1 if missing, else 0).
 - Remove rows/columns only if missing data is very high (> 40-50%).

Summary

We use a Random Forest Regressor for accurate house price prediction because it handles non-linear relationships, is robust, and provides high performance. We preprocess data by scaling features and handling missing values through imputation. Overfitting is detected using validation metrics and learning curves and avoided through regularization, tuning, pruning, and cross-validation.

Example Dataset (Sample)

Location (Encoded)	Size (sq.ft)	No. of Rooms	Amenities (Count)	Age of House (years)	Price (₹ Lakh)
0.35	1200	2	5	10	45.0
0.80	1800	3	8	5	78.0
0.25	1000	2	3	15	38.0
0.60	2400	4	10	2	105.0
0.10	800	1	2	20	25.0
...	...	...	...	...	...

(c) Detecting and Avoiding Overfitting

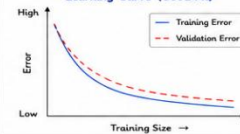
How to Detect Overfitting

- Overfitting occurs when the model learns noise in training data and performs poorly on unseen data.
- Detection methods:
 - Train-Test Split: Large gap between training score and test score indicates overfitting.
 - Cross-Validation: High variance across folds indicates overfitting.
 - Learning Curve: Training error keeps decreasing while validation error starts increasing.
 - Evaluation Metrics on unseen test set (e.g., RMSE, MAE, R^2).

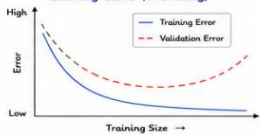
How to Avoid Overfitting

- Use sufficient training data.
- Feature Selection: Remove irrelevant/redundant features.
- Regularization (if using Linear Regression):
 - Ridge (L2), Lasso (L1)
- Hyperparameter Tuning (for Random Forest):
 - Limit `max_depth`
 - Set `min_samples_split`
 - Set `min_samples_leaf`
 - Limit `n_estimators` (avoid too high)
- Use Cross-Validation for model selection.
- Pruning (for Decision Trees).

Learning Curve (Good Fit)



Learning Curve (Overfitting)



Model Evaluation Metrics (Regression)

Metric	Formula	Description
MAE	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Average absolute difference between actual and predicted price.
MSE	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	Average squared difference (penalizes large errors more).
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Square root of MSE; in same units as price.
R^2 Score	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Proportion of variance explained by the model (closer to 1 is better).

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

(a) Describe the working of the K-Means algorithm with a step-by-step example.

(b) Explain how the value of K is chosen using the Elbow Method.

(c) Compare K-Means with Hierarchical Clustering.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

Clustering (Unsupervised Learning)

Clustering is a technique in unsupervised learning that groups similar data points together based on their inherent patterns or similarity, without using any class labels. The goal is to maximize similarity within a cluster and minimize similarity between clusters.

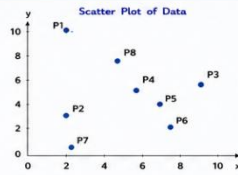
(a) K-Means Algorithm with Step-by-Step Example

K-Means Algorithm (Working)

- Step 1: Choose the number of clusters K.
- Step 2: Randomly initialize K centroids.
- Step 3: Assign each data point to the nearest centroid (based on Euclidean distance).
- Step 4: Recalculate the centroid of each cluster (mean of all points assigned to that cluster).
- Step 5: Repeat Steps 3 and 4 until centroids do not change (convergence).

Example Dataset (2-D Points)

Point	x	y
P1	2	10
P2	2	5
P3	8	4
P4	5	8
P5	7	5
P6	6	4
P7	4	9



Step-by-Step Execution (Let K = 2)

Iteration	Centroids	Cluster Assignment	New Centroids																											
Iteration 1 (Initialization)	Choose K = 2 Randomly select initial centroids: C1(2, 10) (P1) C2(7, 5) (P5)	<table border="1"> <tr><th>Point</th><th>Nearest Centroid</th><th>Cluster</th></tr> <tr><td>P1</td><td>C1</td><td>1</td></tr> <tr><td>P2</td><td>C1</td><td>1</td></tr> <tr><td>P3</td><td>C2</td><td>2</td></tr> <tr><td>P4</td><td>C2</td><td>2</td></tr> <tr><td>P5</td><td>C2</td><td>2</td></tr> <tr><td>P6</td><td>C2</td><td>2</td></tr> <tr><td>P7</td><td>C1</td><td>1</td></tr> <tr><td>P8</td><td>C1</td><td>1</td></tr> </table>	Point	Nearest Centroid	Cluster	P1	C1	1	P2	C1	1	P3	C2	2	P4	C2	2	P5	C2	2	P6	C2	2	P7	C1	1	P8	C1	1	C1 = mean of (P1, P2, P4, P7, P8) $= \left(\frac{2+2+5+4+5}{5}, \frac{10+5+8+9+8}{5} \right)$ $= (2.25, 6.5)$ C2 = mean of (P3, P4, P5, P6) $= \left(\frac{8+5+7+6}{4}, \frac{4+8+5+4}{4} \right)$ $= (6.5, 5.25)$
Point	Nearest Centroid	Cluster																												
P1	C1	1																												
P2	C1	1																												
P3	C2	2																												
P4	C2	2																												
P5	C2	2																												
P6	C2	2																												
P7	C1	1																												
P8	C1	1																												
Iteration 2 Reassign points using new centroids.	Centroids after Iteration 1 C1(2.25, 6.5) C2(6.5, 5.25)	<table border="1"> <tr><th>Point</th><th>Nearest Centroid</th><th>Cluster</th></tr> <tr><td>P1</td><td>C1</td><td>1</td></tr> <tr><td>P2</td><td>C1</td><td>1</td></tr> <tr><td>P3</td><td>C2</td><td>2</td></tr> <tr><td>P4</td><td>C1</td><td>1</td></tr> <tr><td>P5</td><td>C2</td><td>2</td></tr> <tr><td>P6</td><td>C2</td><td>2</td></tr> <tr><td>P7</td><td>C1</td><td>1</td></tr> <tr><td>P8</td><td>C1</td><td>1</td></tr> </table>	Point	Nearest Centroid	Cluster	P1	C1	1	P2	C1	1	P3	C2	2	P4	C1	1	P5	C2	2	P6	C2	2	P7	C1	1	P8	C1	1	C1 = mean of (P1, P2, P4, P7, P8) $= \left(\frac{2+2+5+4+5}{5}, \frac{10+5+8+9+8}{5} \right)$ $= (2.8, 6.8)$ C2 = mean of (P3, P5, P6) $= \left(\frac{8+7+6}{3}, \frac{4+5+4}{3} \right)$ $= (7, 4.33)$
Point	Nearest Centroid	Cluster																												
P1	C1	1																												
P2	C1	1																												
P3	C2	2																												
P4	C1	1																												
P5	C2	2																												
P6	C2	2																												
P7	C1	1																												
P8	C1	1																												
Iteration 3 Reassign again - no change in clusters.	Centroids after Iteration 2 C1(2.8, 6.8) C2(7, 4.33)	<table border="1"> <tr><th>Point</th><th>Nearest Centroid</th><th>Cluster</th></tr> <tr><td>P1</td><td>C1</td><td>1</td></tr> <tr><td>P2</td><td>C1</td><td>1</td></tr> <tr><td>P3</td><td>C2</td><td>2</td></tr> <tr><td>P4</td><td>C1</td><td>1</td></tr> <tr><td>P5</td><td>C2</td><td>2</td></tr> <tr><td>P6</td><td>C2</td><td>2</td></tr> <tr><td>P7</td><td>C1</td><td>1</td></tr> <tr><td>P8</td><td>C1</td><td>1</td></tr> </table>	Point	Nearest Centroid	Cluster	P1	C1	1	P2	C1	1	P3	C2	2	P4	C1	1	P5	C2	2	P6	C2	2	P7	C1	1	P8	C1	1	No change: C1 = (2.8, 6.8) C2 = (7, 4.33) Algorithm converges. These are the final clusters.
Point	Nearest Centroid	Cluster																												
P1	C1	1																												
P2	C1	1																												
P3	C2	2																												
P4	C1	1																												
P5	C2	2																												
P6	C2	2																												
P7	C1	1																												
P8	C1	1																												

Final Clusters:

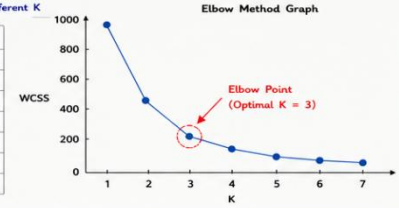
Cluster 1 → {P1, P2, P4, P7, P8} with centroid (2.8, 6.8)
 Cluster 2 → {P3, P5, P6} with centroid (7, 4.33)

(b) Choosing the Value of K using the Elbow Method

Elbow Method: Run K-Means for different values of K and compute Within-Cluster Sum of Squares (WCSS). Choose K at the "elbow" point where the rate of decrease sharply changes.

Example: WCSS for different K

K	WCSS
1	980
2	452
3	235
4	143
5	98
6	72
7	61



After K = 3, the reduction in WCSS is small → hence K = 3 is optimal.

(c) Comparison: K-Means vs Hierarchical Clustering

Aspect	K-Means Clustering	Hierarchical Clustering
Type	Partition-based	Hierarchy-based
Number of clusters	Must be specified (K)	Not required in advance
Output	Single flat clustering	Dendrogram (multi-level clusters)
Working	Start with K centroids, assign points & update iteratively	Builds a tree of clusters either bottom-up (agglomerative) or top-down (divisive)
Distance metric	Usually Euclidean distance	Any distance/similarity metric
Scalability	Efficient for large datasets (O(nkt))	Computationally expensive (O(n²) to O(n³))
Cluster shape	Works well for spherical, compact clusters	Can capture any shape clusters
Handling noise/outliers	Sensitive to outliers	More robust (with suitable linkage)
Flexibility	Less flexible	More flexible
Interpretability	Easy to interpret	Dendrogram gives more insight
Examples	Customer segmentation, image compression	Gene clustering, document clustering, taxonomy

Key Takeaway

- Clustering groups similar data without labels.
- K-Means is simple, fast, and effective for large datasets with compact clusters.
- The Elbow Method helps choose an appropriate number of clusters K.
- Hierarchical clustering provides a hierarchy of clusters and works well for small datasets and complex structures.